AS Level Computer Science Revision Document Amending Suggestion

2025 Seires Mark Scheme Update

1. ~~Page 1.1.3 Perform binary addition and subtraction~~
    a. Overflow
        i. The result if a larger number than can be stored in the given number of bits
        ii. The answer cannot be represented in the number of bits available(w25_11)

2. ~~Page 33 2.1.8 hardware used to support LAN~~
    a. ~~Switch添加further description e.g. (s25_12)~~
        i. receive packets from devices
        ii. forward to intended recipient
        iii. store MAC addresses

3. ~~Page 41 2.1.13 hardware used to support the internet~~
    a. ~~Cell phone network (s25_12)~~
        i. land is spilt into cells for maximum line of sight
        ii. each cell has a tower with an antenna which receives and transmits data
        iii. data is transmitted between the tower and the phone
        iv. data transmission is wireless using low power radio signals
        v. multiple devices can communicate simultaneously with the same tower
    b. ~~Dedicated lines(s25_11)~~
        i. Used to provide a direct / private connection which therefore provides faster transmission

4. ~~Page 21 1.3.3 how a text file, bitmap image, vector graphic and sound file can be compressed~~
    a. ~~Justify lossy(s25_11)~~
        i. reduce more file size than lossless
        ii. …. so significantly less bandwidth
        iii. …. so buffering is reduced

5. ~~Page 15 1.2.6 how sound is represented and encoded~~
    a. analogue data(s25_13)
        i. data values that are continuously changing // variable // any value

        ii. the type of sound wave before it is recorded by a computer

6. ~~Page 122 8.1.3 Show understanding of and use the terminology associated with a relational database model.~~
    a. Referential integrity
        i. 添加Referential integrity ensures that every foreign key has a corresponding primary key到2nd bullet point前面作为conclusion
        ii. 2nd bullet point添加例子: e.g. … is a foreign key in ‘foreign’ table and is dependent on the primary key in ‘primary’ table(w25_11)
7. ~~Page 62 3.2.6 Construct a logic expression~~
    a. Add method to construct a logic expression:
        i. Extract conditions that output 1
    b. 添加两道例题

(b) Write the logic expression for the following truth table.

R	S	T	Q
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

        i. Q = ... [2]

1 mark per bullet point, **max 2** marks

- One correct term: (R AND S AND NOT T) // (NOT R AND NOT S AND T)
- Second correct term **plus** the OR in correct place

Answer:
Q = (R AND S AND NOT T) OR (NOT R AND NOT S AND T)

8. ~~Page 83 5.1.2 Explain the key management tasks carried out by the Operating System~~
    a. Hardware management
        i. Install device drivers (new key word from w25_11)
        ii. Manages communication between devices
        iii. Manages hardware interrupts (new key word from w25_11)

9. ~~Page 95 5.2.4 Describe features found in a typical Integrated Development Environment (IDE)~~
    a. Presentation
        i. Auto-indentation

10. ~~Page 65 4.1.2 Show understanding of the purpose and role of registers, including the difference between general purpose and special purpose register~~
    a. The Accumulator (ACC)
        i. stores the intermediate results of arithmetic and logical operations // holds the result of a calculation

11. ~~Page 68 4.1.5 Show understanding of how factors contribute to the performance of the computer system~~
    a. Why two cores may not run two times faster than one
        i. Multiple cores introduce additional overheads添加alternative sentence
        ii. i.e. Latency may be increased
12. ~~Page 45 2.1.14 Explain the use of IP addresses in the transmission of data over the Internet~~
    a. Static IP address
        i. IP address that does not change each time a device **connects to the network** (w25_11)

13. ~~Page 9 1.2.2 Perform calculations to estimate the file size for a bitmap image~~
    a. Explain why the actual file size may be larger than the one calculated
        i. The file will contain metadata
        ii. The file will have a header

14. ~~Page 128 8.2.2 software tools found within DBMS~~
    a. Developer interface
        i. Explain how make use of // purpose of [3]
            1. To create / modify / delete tables
            2. To create a form for data input
            3. To add a menu to enable users to choose different actions
            4. To add tools to a form
            5. for example, drop-down boxes / buttons etc.
            6. To modify relationships

15. ~~Page 142 10.4.2 show understanding of queue~~
    a. queue
        i. Key features[3]
            1. each queue element contains one data item
            2. two pointer needed to indicate the start and the end of the queue
            3. works on a FIFO basis
            4. may be circular
            5. manipulate by using enqueue()/dequeue [s25_23]

16. ~~Page 159 12.3.4 show understanding of methods of testing available~~
    a. walkthrough[3] [s25_23]
        i. going through the program one line at a time
        ii. using a trace table
        iii. draw up test data, e.g. boundary, normal, abnormal data
        iv. errors will be indicated when the variable gives an unexpected value
        v. faults in logic of the program can be detected

17. ~~Page 100 6.1.3 security measures~~

a. Encryption[2~3]
    i. data is turned into cypher text
    ii. … using an encryption key
    iii. so that it is cannot be understood if intercepted without the decryption key
    iv. by giving example, e.g. VPN
b. Physical method[s25_13]
    i. For example, the computer storing the data cannot be accessed without the key to the room

~~18. Page 102 6.1.4 pharming~~
a. describe[1] [w25_11]
    i. Malicious code/software installed on a computer which redirects user to a fake website to obtain personal data
    ii. Users are directed to a bogus website that looks legitimate to obtain personal information

~~19. Page 76 4.2.4 instructions are grouped~~
a. Loading data into the accumulator is an example of an instruction in the **Data movement** group. Incrementing the index register is an example of an instruction in the **Arithmetic operations** group. Branching to another address is an example of an instruction in the **Conditional and unconditional (jump) instructions** group.

Other Suggestions

1. ~~文档结尾附上往年pastpaper, mark scheme, examiner report, topical question, syllabus, pseudocode guide(前面已有)的google drive链接~~
    a. AS Level Computer Science 9618

2. ~~Page 6 1.1.3 binary addition~~
    a. 添加overflow情况 e.g. (1)1100 0010
    b. 添加largest & smallest two's complement binary number
        i. i.e. largest: 0111 1111
        ii. smallest: 1000 0000

3. ~~Page 67 4.1.4 address bus, data bus, control bus添加~~
Explain how the CU, system clock and control bus operate to transfer data between the components of the computer system.
    a. The system clock gives out timing signals
    b. … which are sent on the control bus
    c. …to synchronise the other system components
    d. The Control Unit initiates data transfer
    e. …by generating signals that are sent on the control bus to other components

4. ~~Page 72 4.1.7 F-E Stage 修正第四个Register transfer notation(w22_12_T7)~~
    a. CIR <- [MDR]

5. ~~Page 72 4.1.7 Fetch stage description添加~~
    a. the instruction is copied to the MDR
    b. … from the address held in the MAR
    c. …which has been read / written [new key word in w25_13]

6. ~~Page74 & 75 4.2.2 two-pass assembler添加对first-pass & second-pass的 description(w22_11_T6)~~
    a. first-pass
        i. Read the assembly language program one line at a time.
        ii. Remove comments.
        iii. Check the opcode is in the instruction set. // To create a symbol table
    b. second-pass
        i. Read the assembly language program one line at a time.
        ii. Generate the object code
        iii. Jump instructions access memory addresses via table
        iv. save or execute the program, error repeated if they exist

7. ~~Page 78 4.2.5 show understanding of and be able to use different modes of addressing~~
    a. Relative addressing
        i. To allow for re-locatable code
        ii. ... because all (target) addresses can be specified by the base address +offset

8. ~~Page 80 4.3.1 logical, arithmetic shifts添加exmaple[s25_13]~~
    a. logical shift
        i. two's complement number 11001010 after a logical left shift of 2 places
        ii. 0010 1000
    b. arithmetic shift
        i. two's complement number 10011110 after arithmetic right shift of 3 places
        ii. 1111 0011

9. ~~Page 80 4.3.1 binary shift 添加e.g. 对LSR#3的解释~~
    a. Identify the mathematical operation that the following instruction will perform on the contents of the accumulator
        i. The number is divided by 8(and only whole number retained)

10. ~~Page 153 12.1.1 purpose of a development life cycle添加(s25_22)~~
    a. Benefit // Purpose // Need of program development lifecycle

        i. easier to plan budget/time
        ii. clear deliverables // prototypes produced at (end of) each stage

11. ~~Page 160 12.3.4 methods of testing available添加**replace**~~
    a. Stub testing[2~3]
        i. **Simple** modules are written to **replace** each of the unfinished modules.
        ii. Each simple module will return an expected value
        iii. test the program before all subroutines have been implemented

12. ~~Page 161 12.3.7 Adaptive maintenance添加description(s25_22)~~
    a. Adaptive
        i. Describe[3]
            1. Change to (based on question) requirements
            2. New technology available
            3. Changes to the program code
            4. Changes in relevant legislation

13. ~~Stack's Implementation添加对pop与push的further descriptions i.e.~~
    a. push: add element to the top of the stack
    b. pop: removes the top element from the stack

14. Page 142 10.4.2 Show understanding that a stack, queue and linked list are examples of ADTs
    a. 缺少Stack的features(待定)

15. Page 155 12.1.3 Describe the principles, benefits and drawbacks of each type of life cycle
    a. 缺少iterative的life cycle description(待定)

16. ~~Page 140 10.3.1 Files~~
    ~~a. 添加meaningful format of the filename(不在syllabus中，但是出现在w25_23，并且根据examiner report，very few candidates answer)~~
        i. include suffix based on <based on the question>
        ii. concatenate with a root filename such as <based on the question>

17. ~~Page 141 Write pseudocode to handle text files that consist of one or more lines~~
    a. 添加benefit and drawback of storing data on a separate line
        i. benefit: algorithm to store/extract a record e.g. is easier
        ii. drawback: more file accesses data are needed

18. ~~Page 148 11.2.1.3 count-controlled loop~~
    a. 添加BREAK语法，pseudocode guide里没有，但是w25_23_T8的MS出现了，并且在examiner report中也有关于BREAK语法的描述

19. ~~Page 160 12.3.5 Show understanding of the need for a test strategy and test plan and their likely contents~~
    a. 添加例题9618_w25_qp_23 T6
    b. 添加description[2]
        i. create a list of e.g. numbers that should generate all of the possible generated values(based on the question)
        ii. … and check each generated value as expected
        iii. by giving examples

20. ~~Page 146 11.1.3 Use built-in functions and library routines~~
    a. 添加advantange of using built-in function // **library routines**
        i. Saves development time / no need to write it / can't write it…
        ii. Pre-compiled and tested / Increased reliability / reduces chance of error
        iii. Available to all programs
        iv. no maintenance of function//routine is needed
21. ~~Page 139 10.2.1 Use the technical terms associated with arrays~~
    a. 添加why sth e.g. -1 is chosen as initial array index
        i. invalid as array index

22. ~~Page140 10.3.1 Show understanding of why files are needed [1]~~
    a. [1]改为[2] (w25_21)
        i. Information is saved after the program ends // after the computer switched off
        ii. the amount of values stored in a text file is not fixed

23. ~~Page 83 5.1.2 key management tasks~~
    a. process management supports multi-tasking[3][s24_12]
        i. schduling processes
        ii. decide which process to be run next
        iii. allow one process to be paused whilst another process can be actioned
        iv. switch between processes to allow them to share the use of
    b. memory management supports multi-tasking[3]
        i. store data from currently running program to RAM
        ii. decide which process should be in main memory
        iii. make efficient use of memory
        iv. Stops the data from overwriting each other in RAM
    c. *basically there are three common points:*
        i. *prevent process using the same memory // overwriting each other in RAM // require same resource*
        ii. *schedule / decide which process to be run next (process)*
        iii. *decide which process should be in main memory (memory)*

24. ~~Page 61 3.1.8 monitoring and control system~~
    a. difference between monitoring and control system

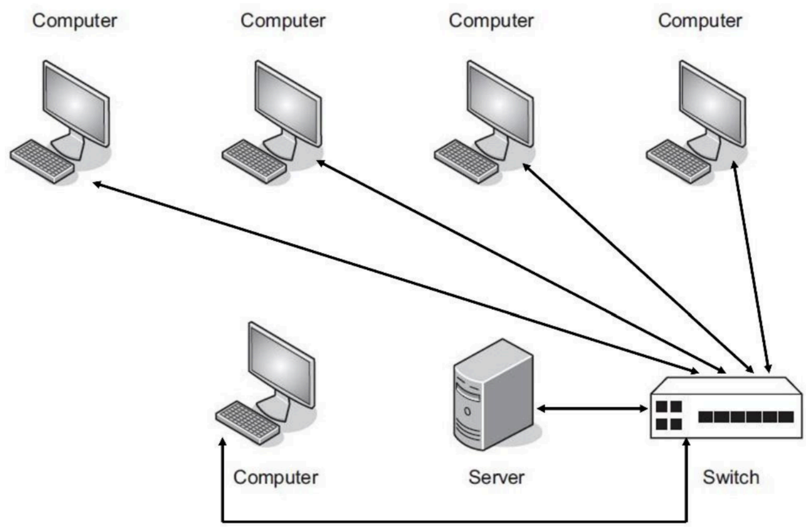
i. control system uses feedback, while monitoring system doesn't use
ii. control system produces action to autonomously change the environment if the values are out of prescribed range[w25_11]
iii. control systems use actuators monitoring systems do not have any actuators
iv. The output from a monitoring system does not affect the subsequent input whereas output from a control system affects the next input.

~~25. Page 34 2.1.8 hardware that used to support a LAN~~

a. 添加示意图 i.e.

1 mark per bullet point, **max 2** marks

- All computers directly connected to switch and no other connections
- Switch connected directly to server



~~26. Page 27 2.1.5 topologies~~

a. Star topology
    i. how data is transmitted [3] [w25_12]
        1. sending device sends packet/data to the router/switch
        2. the switch/router checks the destination of each packet
        3. … using routing table
        4. sends them **only** to the intended receiver

~~27. Page 112 different types of software licensing~~

~~a. Open Source (Initiative)~~
    i. describe [1]
        1. Software can be freely copied, distributed or adapted
        2. *better define the term with the how the software can be used i.e. copied, distributed, adapted*

~~b. Software License~~
    i. describe [1]

            1. A (legal) agreement that defines the rights and terms of usage for the software

~~28. Page 90 5.2.1 need for interpreter and compiler~~

  a. Interpreter添加 ii点
     i. using while writing the program
     ii. … to test/debug the partially completed program
     iii. … because errors can be corrected in real-time

~~29. Page 114 7.1.4 different types of software licensing~~

  a. Free Software Foundation
     i. Describe [2]
        1. is free of restrictions
        2. not necessarily free of charge
        3. a user can edit <u>source code</u>
        4. user can share software with others
        5. *note that it is not necessarily free of charge*

Syntax & Grammar Suggestions

~~1. Page 26 thin-client and thick-client~~

  a. Thin-client description
     i. server perform**<u>s</u>** ….
     ii. clients only send ….

~~2. Page 31 implications of the use of wireless and wired networks~~

  a. comparing copper cable and fire-optic cable
     i. fibre-optic cable ha**<u>s</u>** ….
     ii. fibre-optic cable need**<u>s</u>** ….

~~3. Page 69 4.1.5 factors contribute to the performance of computer system~~

  a. Cache
     i. prevents CPU from **<u>idling</u>** while waiting for data

~~4. Page 74 4.2.2 Describe the different stages of the assembly process for a two-pass assembler~~

  a. First pass的Describe部分缺少分值 i.e. [2]